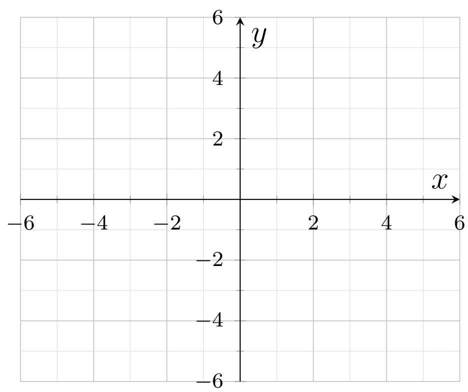
Solving Systems of Two Linear Equations by Graphing

Graphing two lines from slope-intercept form, graphing from standard form, recognizing parallel and identical lines, and reading a real-world intersection.

- Graph *both* lines on the same grid. The solution of the system is the point where they cross, written as an ordered pair (x, y).
- From $y = mx + b$: plot the y-intercept b first, then use the slope m as rise over run to step to a second point.
- From standard form $Ax + By = C$: the two intercepts are quickest. Set $x = 0$ to find one, then $y = 0$ to find the other.
- Always check your point by substituting it back into *both* equations.
- Two lines that never cross have *no solution*; two lines that lie on top of each other have *infinitely many.*

1. Graph both lines and write the solution as an ordered pair.

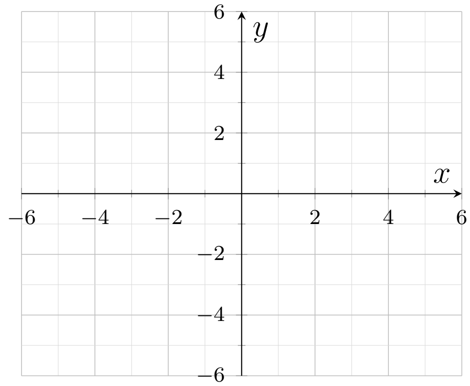
$$y = x + 2 \qquad \text{and} \qquad y = -x + 6$$



Answer: ________________

2. Graph both lines and write the solution as an ordered pair.

$$y = 2x - 3 \qquad \text{and} \qquad y = x + 1$$



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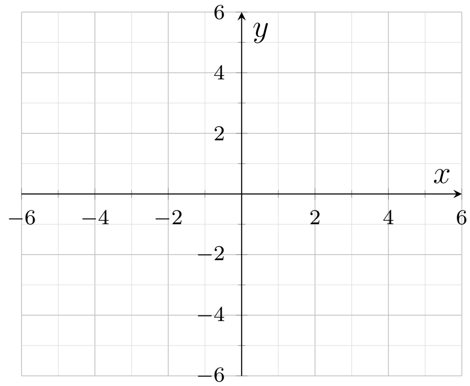
3. Graph both lines and write the solution as an ordered pair. One line falls and one rises — say which is which before you graph.

$$y = -2x + 7 \qquad \text{and} \qquad y = x - 2$$



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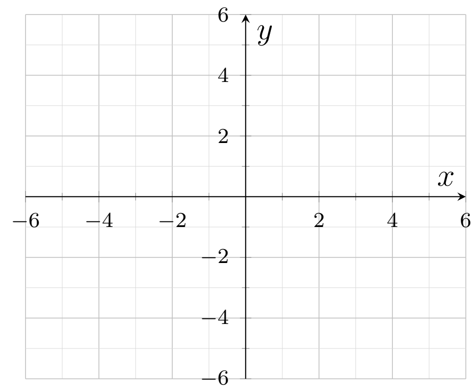
4. Both equations are in standard form. Find the intercepts of each line, graph them, and write the solution as an ordered pair.

$$x + y = 5 \qquad \text{and} \qquad x - y = 1$$



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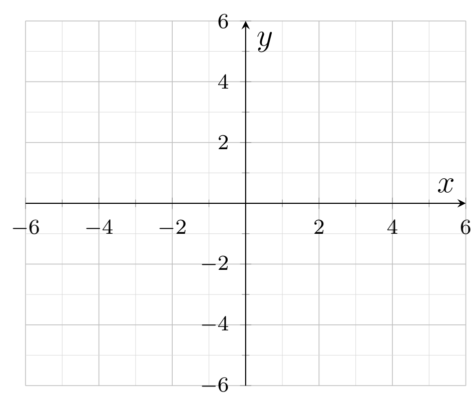
5. One slope is a fraction, so count the rise and run carefully. Graph both lines and write the solution as an ordered pair.

$$y = \tfrac{1}{2}x + 1 \qquad \text{and} \qquad y = -x + 4$$



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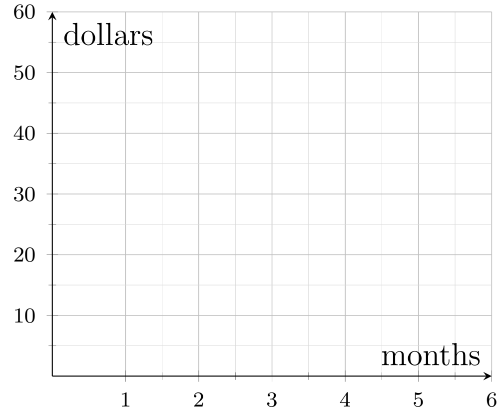
6. Two clubs charge a joining fee plus a monthly fee. Club A costs 20 dollars to join and 5 dollars a month, so the total after x months is $y = 5x + 20$. Club B costs 10 dollars to join and 10 dollars a month, so its total is $y = 10x + 10$. Graph both totals and write the ordered pair where the clubs cost the same.



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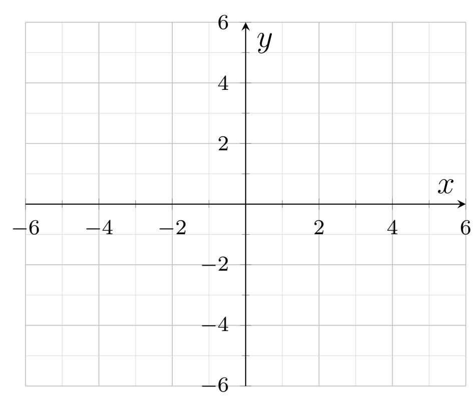
7. Graph both lines from their intercepts and write the solution as an ordered pair.

$$2x + y = 8 \qquad \text{and} \qquad x - y = 4$$



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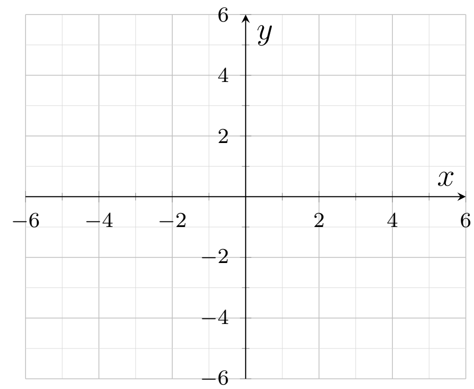
8. Graph both lines. Then answer: how many ordered pairs solve this system, and how does the graph tell you?

$$y = 3x + 1 \qquad \text{and} \qquad y = 3x - 2$$



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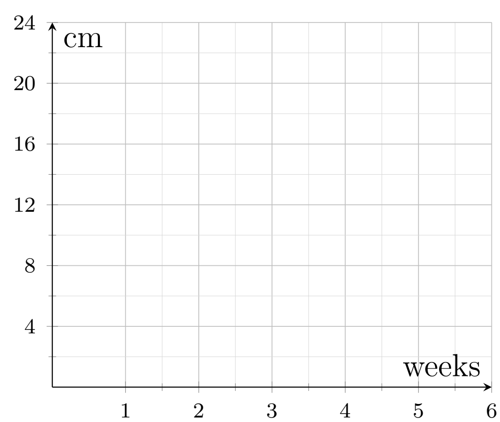
9. Two bean plants are measured each week. Plant A is 6 cm tall and grows 4 cm a week, so its height after x weeks is $y = 4x + 6$. Plant B is 2 cm tall and grows 6 cm a week, so its height is $y = 6x + 2$. Graph both heights and write the ordered pair where the plants are the same height.



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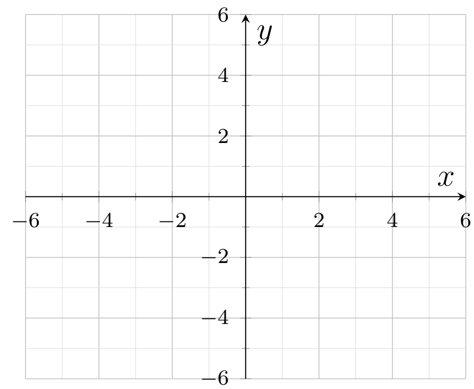
10. Rearrange each equation into slope-intercept form first, then graph both lines and write the solution as an ordered pair.

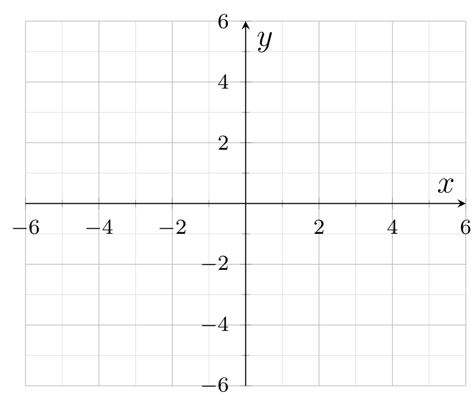
$$3x - y = 5 \qquad \text{and} \qquad x + y = 7$$



Answer: ____________________

11. Graph both lines. Then answer: how many ordered pairs solve this system, and what do you notice about the two equations?

$$2x + 2y = 8 \qquad \text{and} \qquad y = -x + 4$$



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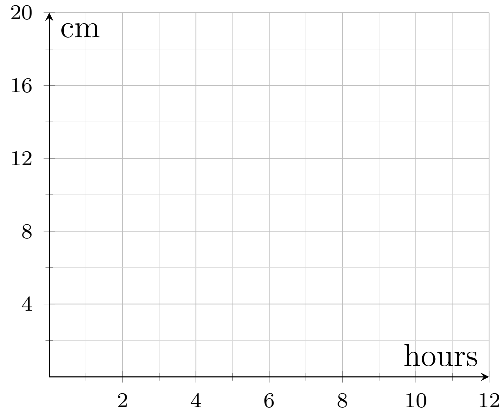
12. Two candles are lit at the same time. The tall candle is 20 cm long and burns 2 cm an hour, so its height after x hours is $y = -2x + 20$. The short candle is 12 cm long and burns 1 cm an hour, so its height is $y = -x + 12$. Graph both heights, write the ordered pair where the candles are the same height, and state which candle is taller before that time.



Answer: ____________________